

Complex landscapes of spatial interaction

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Abstract. How complex is the spatial economy? Some apostles of complexity argue that complex behaviour can arise in any system consisting of a largish number of intelligent, adaptive agents interacting on the basis of local information only. This paper examines several features of such dynamic systems, including path-dependence, emergence and self-organization. It goes on to explore their importance for the spatial sciences. Because space scales can change abruptly from local to global, strongly-interactive spatial economies sometimes exhibit astonishing collective properties, emergent features which are lawful in their own right. Segregation, self-similarity and the rank-size rule are familiar examples. To understand how collective order arises from seemingly random fluctuations, we must note how agents choose to interact with other agents and with their environment. We must synthesize rather than analyse. In the paper, self-organization is explored in a variety of contexts, including Schelling's model of neighborhood segregation and some work with cellular automata that has sharpened our insights into the collective synthesis of agents' interactions. Power laws are widely observed. A new way of doing social science – agent-based simulation – offers powerful new insights. It seems likely to revolutionize our field, along with the whole of the social sciences. Some of the current research underway in this area is discussed.

“Truth is much too complicated to allow anything but approximations.”

JOHN VON NEUMANN

1. Introduction

Recently, Paul Krugman (1994) posed the intriguing question: “How complex is the economic landscape?” Fondly enough, he was not thinking of mountains and rivers over which goods are transported and services channelled, but of an *abstract* landscape, one that represents the dynamics of resource allocation across activities and locations. Given his longstanding interest in international trade, Krugman pointed to some issues that add complexity to the landscape of trade: increasing returns, the prevalence of multiple equilibria, and the importance of path-dependence, just to name a few.

His conclusion was that the metaphor of complex landscapes should be added to the economists' vocabulary for two reasons. First, it is a suggestive image that may help to focus economists' work. Second, the idea of dynamics over complex landscapes is central to the growing interdisciplinary literature on *complexity*. After noting that a handful of economists have been dabbling in this arena for some time, he suggested that more economists should recognize a promising opportunity for two-way trade in ideas. They could learn from the apostles of complexity, and these visionaries could also learn something from economists.

In this paper, I shall argue that the metaphor of complex landscapes should also be added to the vocabulary of urban and regional scientists.¹ I offer three reasons. First, there is an unresolved debate about the respective roles of necessity and chance in patterns of industrial development over space.² Those backing necessity see industrial location as *deterministic*, while those backing chance see it as *path-dependent*. In other words, this debate pits simple against complex.

Second, complexologists argue that phase transitions can transform simple socio-economic systems into complex ones, and vice versa. Such transitions are highly sensitive to the spatial scale of the interactions between the agents involved. Because space scales can change abruptly from local to global, strongly-interactive spatial economies sometimes exhibit astonishing collective outcomes, emergent properties that are lawful in their own right. Cities and regions can *self-organize*. To understand the extraordinary macroscopic pattern and structure that cities and systems of cities can embrace, we must look into the nonlinear character of agents' interactions.

Third, unravelling the behaviour of large numbers of agents – and their patterns of interaction in geographical space – quickly gets too difficult for a meaningful mathematical solution. Most of the closed-form models familiar to regional scientists, as well as those being advocated by some “new economic geographers,” are incapable of coming to grips with phenomena like emergence and self-organization. In such uncertain environments, a key research tool to help us unravel the complexity is *agent-based simulation*. These simulation experiments inside the computer involve three basic ingredients: agents, an environment or space, and a set of behavioural rules. Just a few simple rules can produce richly complex outcomes. Agent-based simulation represents another way of doing social science. Like deduction, it starts with an explicit set of assumptions. Unlike deduction, it does not prove theorems. Instead it generates simulated data that can be analyzed inductively. Its real purpose is to sharpen our *intuition* about patterns and processes that shape the economic landscape over time and space.

In the next section, the twin forces of chance and necessity are discussed in

¹ Most of the topics developed in this paper are discussed at greater length in Batten (2000).

² In the urban planning literature, necessity permeates those planning models and analyses that rely on principles of optimization and equilibrium. These methods do not deal with the means to an end, but assume an end in itself. The Lowry model is an influential example. Although Lowry demonstrated that a large-scale metropolis could be “simulated” inside a computer, mimicking the locational choices of households and the growth of retail centres in 650 mile-square tracts of the Pittsburgh region, his model was really nothing more than a vast system of assumptions with no identification of the processes which might lead to such a solution. In other words, people could not “mix and sort” themselves in ways that they have been observed to do in cities and regions.

the context of industrial development. Both sets of forces play a powerful role in shaping the landscape of the spatial economy. Section 3 looks at self-organization in the form of Schelling's (1978) deceptively simple account of how neighborhoods in a city could become segregated. This classic attempt at agent-based modelling shows why distinctions between weakly-interactive and strongly-interactive patterns of interaction are important in the spatial economy. If agents are sensitive to particular patterns – as they are under Schelling's regime – a qualitatively different kind of collective behaviour can arise. This is the hallmark of emergence.

Comparisons of weakly- and strongly-interactive systems are extended to intercity networks in Sect. 4. A simple toy example, based on random graph theory, reveals that the abrupt change from a weakly- to a strongly-interconnected network resembles a phase transition. Some of the collective behaviour on a spatial network may mimic Per Bak's (1994) self-organizing sandpile model.³ For example, a state of self-organized criticality can occur on traffic networks. In light traffic, interactions are mostly local. Once vehicles become more strongly dependent on neighboring vehicles, however, the interdependencies can become global. Suddenly, stop-start waves appear and the risk of heavy congestion follows. In this dynamic world, avalanches of change obey simple power laws. The pervasiveness of power laws, and their significance in an urban and regional context, are examined in Sects. 5 and 6.

Some emergent properties of a spatial economy are surprising because it is difficult to anticipate the full consequences of seemingly simple interactions between agents. In Sects. 7 and 8, I argue that our intuition about emergence can be sharpened through the lens of agent-based simulation. Simulations based on cellular automata (CA) have several advantages over micro-simulation techniques.⁴ Importantly, they link macrostructure to microbehaviour and they are explicitly dynamic. In Sect. 7, two-dimensional CA are defined and a few urban simulation models based on CA are discussed. Perhaps the greatest attraction of a CA-based approach for the spatial sciences is the equal weight given to the importance of space, time and system attributes.

An important challenge for agent-based simulation is to represent the interface between changes at the human and physical levels. Agents can change their behaviour generators very quickly, especially in response to the moves of other agents, whereas developmental changes across networks of settlements and infrastructure can be relatively slow to adjust. In Sect. 8, we look at a silicon landscape invented by Joshua Epstein and Robert Axtell (1996). Known as *Sugarscape*, it's an ever-changing environment where a set of agents interact with each other according to a set of rules. Even with relatively simple rules, fascinating things happen once the agents start to interact. *Sugarscape* is an important laboratory because it's crossdisciplinary, process-dependent and recognizes the heterogeneity of human agents. It typifies the

³ Bak's model is important because it may be a mechanism for generating complexity. His claim that real economies behave more like sand, because of discreteness and friction effects, has been explored in detail in Batten (2000).

⁴ Because agent-based simulation is a "bottom-up" approach, it may be contrasted with the "top-down" character of micro-simulation. The former models behaviour as resulting from simple rules, whereas the latter typically relies on equations that are statistically estimated from aggregate data.

latest way of doing social science, as we enter the unprecedented era of CA and agent-based simulation. Section 9 contains a few thoughts on the possible ramifications of this new era for the spatial sciences.

2. Chance and necessity in spatial development

Brian Arthur (1994) suggests that the literature on industrial location contains two different world-views.⁵ First there are the writings of von Thünen, Weber, Predöhl, Christaller and Lösch, all of whom saw the spatial evolution of industry as *preordained* – by geographical endowments, transport possibilities, and economic needs. In this view, history does not matter. All the key factors are perfectly visible: geographical differences, shipment costs, market interactions, and the spatial distribution of prices and rents. The outcome is determinate and readily predictable: a unique equilibrium pattern. Because this is a static view of how industries locate themselves in geographical space, we shall assign to it the label *stasis*.

Conversely, a second group saw industry location as *path-dependent*. These scholars viewed spatial development as an organic process – with new industry and jobs influenced by, and thus reinforcing, the locational landscape already in place. Included among this group were the later Weber, Engländer, Ritschl and Palander. Although there is still a role for geographical endowments and economic factors – such as transportation costs – in this view, the driving forces are agglomeration economies. Frustratingly, the resulting pattern of industrial location is not unique. A different set of early events can steer a locational pattern into a different outcome. Since the locational system generates structure as it goes, this view is fundamentally dynamic. Possessing a multiplicity of outcomes, the path finally chosen is unpredictable in advance. We shall refer to this view of the locational world as *morphogenesis*.

Stasis or morphogenesis? Necessity or chance? A simple or a complex spatial economy? Which explanation is correct? Many of the spatial patterns we observe today have been forged by a mixture of chance and necessity, rather than by either factor alone (Allen and Sanglier 1981; Krugman 1993). In the world of morphogenesis, however, lie the seeds of unexpected change (see Batten 1982). Whether chance events in history matter in determining patterns of human settlement, growth and change reduces to a question of topology. It hinges on whether the underlying structure of locational forces guiding the location pattern is convex or nonconvex (Allen and Sanglier 1981; Arthur 1994). History does matter when these forces are nonconvex, and nonconvexity stems from some form of agglomeration or increasing returns in space.

Agglomeration is a powerful force. For example, firms that are not heavily reliant on raw material locations, but are more sensitive to their industry's learning curve, are often attracted to a large metropolis by the presence of other like-minded firms. Some densely populated cities can offer better infra-

⁵ The seeds of this debate were sown by several members of the great German industry location school. As Isard (1956) noted, Engländer and Palander were severe critics of Weber's early theory, claiming that he grossly underemphasized the actual development process and the historical advantages of existing production points as self-reinforcing centres of agglomeration. For a discussion of these conflicting views, see Isard (1956, Chap. 8) or Arthur (1994, Chaps. 4 and 6).

Table 1. Two spatial worlds – necessity and chance

Necessity A simple world	Chance A complex world
Stasis	Morphogenesis
Unique outcome	Multiple outcomes
Equilibrium	Path-dependent
Mechanistic	Organic and selective
Predictable	Unpredictable
Diminishing returns	Increasing returns
Convex	Nonconvex
Easy to model	Difficult to model

structure, more diverse labour markets, more specialized services, and more opportunities to do business face-to-face. They may also provide an active forum for the continuous exchange of ideas. This can set off a chain reaction, convincing more and more firms to locate in the same place. Under such conditions, the world of morphogenesis dominates. It's in this dynamic world that we can find nonequilibrium processes like self-organization.

3. The self-organizing economy

To see the process of self-organization at work, we can turn to the pioneering attempts at agent-based modeling carried out a generation ago by Thomas Schelling (1969, 1978).⁶ His classic ideas on complexity and self-organization were summed up in a deceptively simple account of how neighborhoods in a city could become segregated. In Schelling's model, there are two classes of agents. He thought of them as blacks and whites, but they could be any two classes of individuals that have some difficulty in getting along together – e.g. boys and girls, smokers and non-smokers, butchers and vegetarian restaurants. For our purpose, a chessboard can play the role of our “simplified city”. Think of the sixty-four squares as a grid of potential house locations, although the principles hold just as convincingly over much larger (and irregularly-shaped) spatial domains.

Each agent cares about the class of his immediate neighbors, defined as the occupants of the abutting eight squares of the chessboard. Although there are eight squares abutting each square, less than eight of the squares may be filled with neighbors. Preferences are honed more by a fear of being isolated rather than from a liking for neighbors of the same class. It's pretty obvious that such preferences will lead to a segregated city if each agent demands that a majority of his neighbors be the same class as himself. The novelty of Schelling's work was that he showed how much milder preferences, views that seem compatible with an integrated structure, typically lead to a highly segregated city – once the interdependent nature of any changes are taken into account.

⁶ Krugman suggests that the first chapter of Schelling's (1978) book is “surely the best essay on what economic analysis is about, on the nature of economic reasoning, that has ever been written.” (Krugman 1996, p. 16). The two chapters on “sorting and mixing” provide an excellent, non-mathematical introduction to the idea of self-organization in cities.

	A	B	C	D	E	F	G	H
1		#	○	#	○	#	○	
2	#	○	#	○		○	#	○
3	○	#	○		○	#	○	#
4	#	○		○	#	○	#	○
5	○	#	○	#	○	#	○	#
6	#	○	#	○	#	○	#	○
7	○	#	○	#	○	#	○	#
8		○	#	○	#	○	#	

Fig. 1. An almost-integrated pattern of residential location

Consider the following simple rule: an individual who has one neighbor will only try to move if that neighbor is a different type; one with two neighbors wants at least one of them to be the same type; one with three to five neighbors wants at least two to be his or her type; and one with six to eight neighbors wants at least three of them to be like him or her.⁷ At the individual level, this rule of neighborhood formation is only mildly color-conscious or culture-sensitive. With such preferences it is possible to form an integrated residential pattern that satisfies everybody. The familiar checkerboard layout, where most individuals have four neighbors of each class, does the trick – as long as we leave the corners vacant.

Nobody can move in such a layout, except to a corner. There are no other vacant cells. But nobody wants to move anyway. Since it's an integrated equilibrium structure, there's no incentive to change it. But what if a few people are forced to move. What if three neighbors, who work together at a nearby office, are relocated interstate by their company. They must sell up and move to another place. Will the integrated equilibrium remain? Let's try to find out.

After they leave, the neighborhood layout looks like the board shown in Fig. 1. The departing workmates vacated the squares located at coordinates C4, D3 and E2. Once they move out, however, other nearby neighbors of the same type suddenly feel too isolated. For example, residents at D1 and F1 discover that only one of their four neighbors is the same class as them. Thus they decide to move to locations where the neighborhood rule is satisfied again, say A1 and H8.

A self-reinforcing pattern of interdependencies becomes evident. Another resident can become unhappy because the departing resident tips the balance in his neighborhood too far against his own class, or because his arrival in a new location tips the balance there too far against agents of the other class. Surprisingly, our integrated equilibrium begins to unravel. An unsatisfied individual at C2 moves to C4, leaving another at G2 with nowhere to go. G2 has no alternative but to move out completely, precipitating a chain reaction of moves in response to his decision. Residents at F3, H3, G4, H5, E4, F5 and G6 all follow suit. Despite the fact that agents only have mild preferences against being too much in the minority, some of them are forced to move out

⁷ An equivalent way of stating this rule is that each individual is satisfied as long as at least three-eighths of his or her neighbors are of his or her class.

	A	B	C	D	E	F	G	H
1	#	#	○		○		○	
2	#	○		○		○		○
3	○	#	○		○		○	
4	#	○	#	○		○		○
5	○	#	○	#	○			
6	#	○	#	○	#			○
7	○	#	○	#	○	#	○	#
8		○	#	○	#	○	#	#

Fig. 2. The pattern after the chain reaction of moves

	A	B	C	D	E	F	G	H
1		#	○		○		○	
2	#	○				○	#	○
3	○	#	○				○	
4	#	○		○		○	○	○
5		#	○	#	○		○	#
6	#	○	#	○	#	○		○
7	○		○	#	○	#	○	#
8		○	#		#	○	#	#

Fig. 3. Perturbing the pattern even further

and pockets of segregation begin to appear on our chessboard city (see Fig. 2).⁸

There are now 49 agents residing in the city. Let's trigger some more change by removing another nine of them using a random number generator, then picking five empty squares at random and filling them with a new type of agent on a 50/50 basis. The pattern shown in Fig. 1 is unstable with respect to some random shuffling, and tends to unravel even further. Figure 3 shows the result after our random number generator has done the job.

Some other residents will now be unhappy with their locations and will move (or move again). Seemingly simple moves provoke self-reinforcing responses. Thus a new chain reaction of moves and countermoves is set in motion. To simulate this chain reaction on a computer, the order in which people move, and the way they choose their new location, would need to be specified. As we are doing this by hand on a chessboard, we can watch the structure evolve. When it finally settles down, my randomly generated series of moves leads to the layout in Fig. 4.⁹

What a surprise! Even though the individuals in our city are tolerant enough to accept an integrated pattern, they still end up highly segregated. Even though their concerns are local, the whole city gets reorganized into homogeneous residential zones. Surprisingly, short-range interactions can produce large-scale structure. Our chessboard city has engaged in a process of self-organization. Large-scale order has emerged from a disordered initial state. Segregation may not be our favorite form of order, but it's order nevertheless. It's also meaningful, because all city-dwellers in Fig. 4 are now content.

This large-scale order emerges because the integrated pattern (shown in Fig. 1) is *unstable*. Scramble it a little and you start a chain reaction of moves

⁸ Because of the limited computational power available a generation ago, Schelling discovered pockets of segregation by moving coins around on top of a table decorated with suitable grid paper. As he notes: "Some vivid dynamics can be generated by any reader with a half hour to spare, a roll of pennies, a roll of dimes, a tabletop, a large sheet of paper and a spirit of scientific enquiry or, lacking that spirit, a fondness for games." (see Schelling, 1978, p. 147).

⁹ This layout is one of a number of possibilities, since the order in which individuals move remains unspecified. The final outcome will also be sensitive to the initial conditions (as depicted in Fig. 1). As Schelling noted, repeating the experiment several times will produce slightly different configurations, but an emergent pattern of segregation will be obvious each time.

	A	B	C	D	E	F	G	H
1	#	#	#	#	○	○	○	
2	#	#	#			○	○	○
3	#	#			○	○	○	
4	#	#	#	○	○	○	○	○
5	#	#	○		○		○	
6	#	○		○				○
7	○		○		○		○	#
8		○				○	#	#

Fig. 4. A segregated city

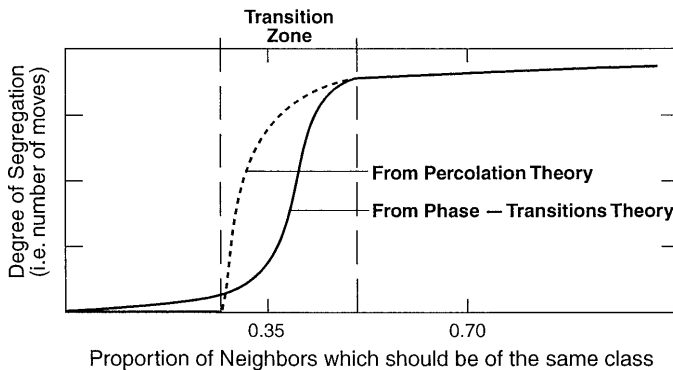


Fig. 5. A phase transition in the degree of segregation as class consciousness changes

that leads to segregation. We could say that you get *order from instability*. This is another of the hallmarks of self-organization.

Schelling fine tuned his rules very carefully. He specified that an individual would be satisfied as long as at least 37.5% of his or her neighbors were of the same class. If that figure had been a fraction lower, say 33.3%, then only two residents in Fig. 1 – those located at positions D1 and F1 – would have wanted to move. Once they had moved – say to A1 and H1 – then everyone else in the city would have been satisfied. In other words, the original integrated equilibrium would have remained stable. There's an important message here about emergence. As Holland (1998) has suggested, emergent properties of agents' interactions are bound up in the selection of rules or mechanisms that specify the model. In this case, a small change in class consciousness – the migration rule – can result in a large change in the ensuing number of moves. There's a small range over which the degree of segregation is by no means obvious. Once class consciousness gets too strong, however, a highly segregated residential pattern appears immediately.

The sudden and unexpected appearance of highly segregated areas is indicative of a qualitative change in the aggregate pattern of behaviour. We might say that the location pattern has “flipped” into an entirely different state. In fact this nonlinear change is indicative of something like a *phase transition*. At first the integrated equilibrium remains rather stable to slight increases in class consciousness. Then, rather suddenly, the number of moves

skyrockets dramatically. Although we cannot be sure that the whole city ever reaches a state of self-organized criticality, various avalanches of change (in the form of clusters of migration of different sizes) can occur. Global order emerges from the growing scale of local interactions.

4. Weakly- and strongly-interactive landscapes

The idea that local interactions can produce global structure – via non-equilibrium phase transitions – came from the pioneering work of some physicists and chemists studying self-organization in physical systems.¹⁰ Yet Schelling's model permits us to see exactly how the process works in a spatial economy. To some extent, of course, the model oversimplifies reality. The tendency is to divide a city into vast $\#$ and 0 areas. What really happens is that the chain reaction of moving households dies out at some point, leaving the city locked into various $\#$ and 0 domains of different sizes. And the resulting classes of individuals are not simply two-dimensional. They're n -dimensional, so much so that it's sometimes difficult to discern the true class or "colors" of all your neighbors.

Despite these drawbacks, Schelling's insights were well ahead of their time and the rich dynamics contained therein are quite extraordinary. Such a rich model helps us to understand why distinctions between weakly-interactive and strongly-interactive patterns of interaction are so important in a spatial economy. The transition zone between these two states provides basic clues about how the behaviour of a group of agents, or even a whole society, may undergo unexpected change. When patterns of interaction between agents become sufficiently dense, as they did in Schelling's model, a qualitatively different kind of collective behaviour arises. Something unexpected happens.

To elaborate further on this kind of transition, a toy problem can be helpful. Toy problems offer insights into more complicated, real-world ones. The problem of interest to us involves random graphs and was conceived originally by Stuart Kauffman (1995). A random graph is similar to a standard graph, except that the nodes are connected *at random* by a set of links. Although we know how the graph or network looks at any time (i.e. which pairs of nodes are already linked), we have no way of knowing which pair of nodes will be linked next. Our concern, then, is with the overall *pattern* of links (i.e. the graph's interconnectivity) rather than the strength of particular links.

¹⁰ The notion of phase transitions has its roots in the physical sciences, but it's relevance to economic evolution has been recognized recently. In the social sciences, phase transitions are difficult to grasp because the *qualitative* changes are hard to see. Far more transparent is the effect of temperature changes on water. As a liquid, water is a state of matter in which the molecules move in all possible directions, mostly without recognizing each other. When we lower its temperature below freezing point, however, it changes to a crystal lattice – a new solid phase of matter. Suddenly, its properties are no longer identical in all directions. The translational symmetry characterizing the liquid has been broken. This type of change is known as an equilibrium phase transition. Recent advances in systems theory, especially studies led by Ilya Prigogine and the Brussels school of thermodynamacists, have discovered a new class of phase transitions – one in which the lowering of temperature is replaced by the progressively intensifying application of non-equilibrium constraints. It's nonequilibrium phase transitions that are associated with the process of self-organization. See, for example, Nicolis and Prigogine (1977, 1989).

This is the starting point for Kauffman's toy problem. Kauffman visualized it in an everyday context, thinking of the nodes as "buttons" and the links as "threads." His idea was to choose any two buttons at random from a pile on the floor, pick them up, and connect them with a thread. If you continue threading buttons in this way, eventually a point is reached where the connectivity of the whole button population changes suddenly.

To put Kauffman's toy problem into the context of a spatial economy, think of the buttons as places – towns and cities – and the threads as transport links – like roads and railroads. Imagine that we randomly choose two places from all the towns and cities inside a country and link them directly by road. Gradually we choose more pairs of places and do the same. As we continue to do this, at first we'll mostly pick pairs of places that have not been linked before. In other words, the linked network of towns and cities features only pairwise road connections. Sooner or later, however, we'll pick a pair of places and find that one of them is already connected to another place. On the whole, however, we could say that the collection of towns and cities is only *weakly-interactive*, in the sense that the spatial scale of interaction possibilities is relatively small.

As we go on choosing pairs of places randomly to link together by road, eventually we find that some towns and cities become interconnected into larger network clusters. For example, as we link the twenty places shown in Fig. 6 using a random number generator, we see that only seven of the twenty places remain outside one particular cluster by the time the ratio of roads to places reaches 0.75 (Fig. 6c).

Denoting the ratio of roads to places by R , the interesting thing is that random graphs exhibit very regular statistical behaviour as one tunes this ratio. Once R passes the 0.5 mark, something unexpected happens. Suddenly most of the clusters become cross-connected into one giant structure! When this giant cluster forms, the majority of places are directly or indirectly connected to each other. As R approaches one, virtually all of the remaining, hitherto isolated places and local clusters become cross-connected into this giant web. Suddenly the whole settlement structure becomes *strongly-interactive*.

This sudden and unexpected change in the size of the largest cluster, as R passes 0.5, is the signature of something akin to a phase transition (see Fig. 5). The size of the largest cluster increases slowly at first, then rapidly, then slows again as R increases further. If there were an infinite number of places, then the size of the largest web would jump discontinuously from tiny to enormous as R passed 0.5. The result is symptomatic of a phase transition, like when separate water molecules freeze to form a block of ice.

This toy example suggests that some of the evolutionary characteristics of the spatial economy may be revealed by viewing it as a *network economy*. If such an economy is only weakly bound together, mostly by local linkages, we may think of it as weakly-interactive. This simple state allows us to analyse its behaviour using simple extrapolation methods or linear models. On the other hand, if many small clusters become cross-linked to form several larger clusters, space scales may suddenly become global. To understand the behaviour of this strongly-interactive economy, other approaches are necessary. Spatial economies tend towards this more complex state over time.

Surprisingly enough, the collective behaviour of a network economy is reminiscent of the self-organizing sandpile model, conceived by Per Bak

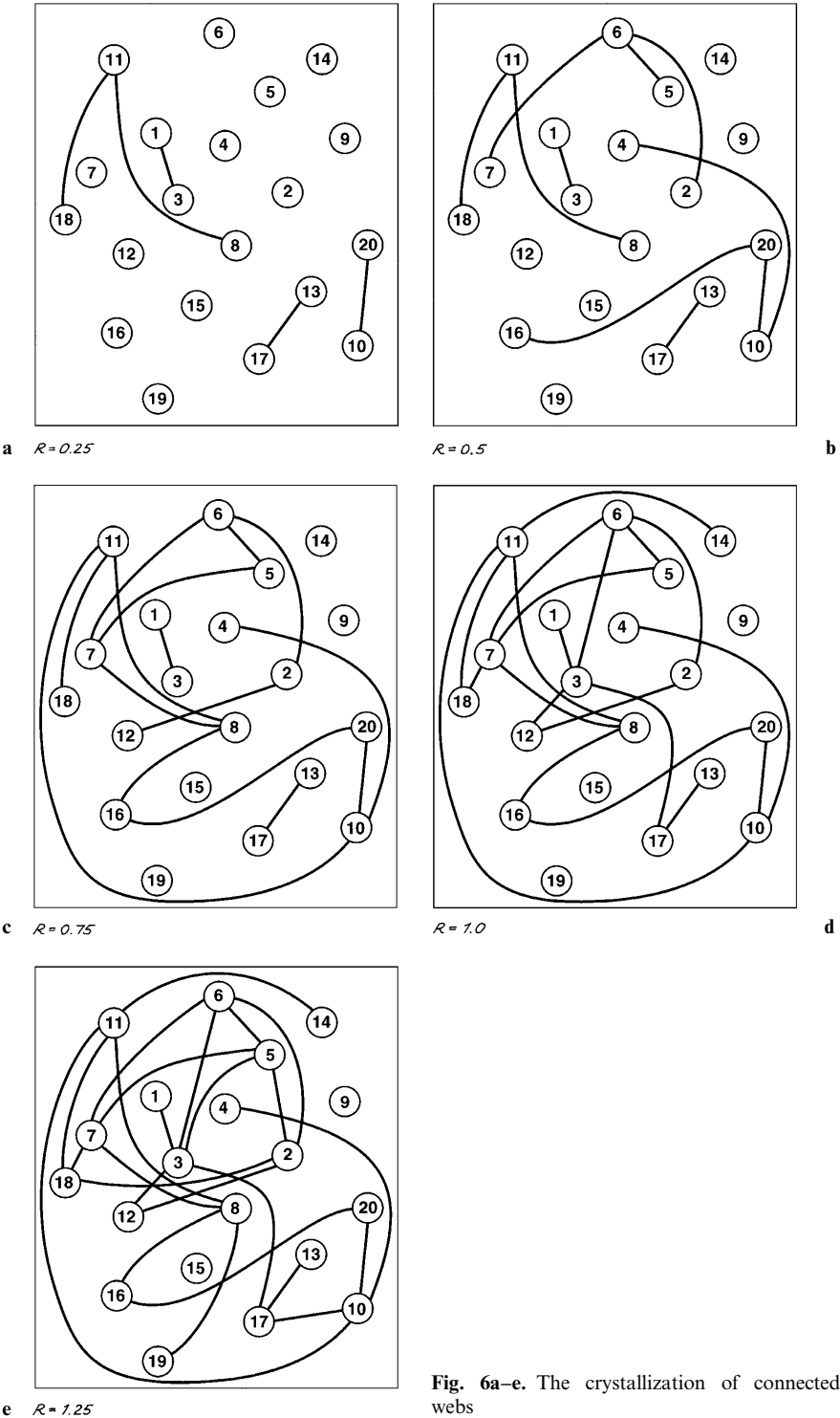


Fig. 6a–e. The crystallization of connected webs

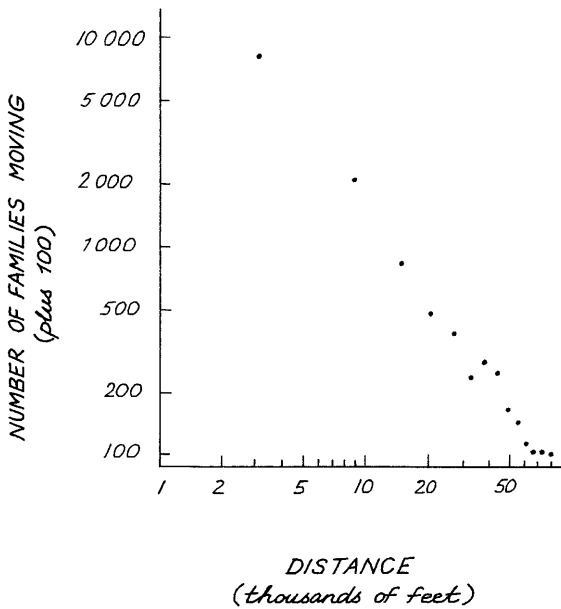


Fig. 7. The number of families moving varying distances in Cleveland is governed by a power law. (Source: Zipf, 1949)

(1996).¹¹ In Bak's sandpile, self-organized criticality is reached once local interactions between separate grains of sand are replaced by global communication throughout the whole sandpile. Avalanches of change follow a simple power law. Self-organized criticality generates this complexity. We'll look more closely at these phenomena in the next section.

5. Power laws and spatial avalanches

There's something very special about those chain reactions of moves – or avalanches of change – that produce and maintain a highly segregated spatial economy. To understand this special property, we need to remember that there's plenty of friction in economies (see Griffin 1998). In a spatial economy, the friction of distance is the classical form of resistance to change. Empirical support for the friction of distance was provided half a decade ago by Zipf (1949). He suggested that the number of families moving between separate areas varied inversely with distance. Figure 7 depicts this relationship.

Human interactions and decision making tend to reinforce such friction effects. Despite its tendency to discourage movement by individuals, it might just be the friction of distance that binds villages, towns and cities together in special patterns to form a stable, dynamic, economy. Interestingly, it's also

¹¹ Bak's model is important because it may be a general mechanism for generating complexity. His claim that real economies behave much more like sand, because of discreteness and friction effects, has been explored in detail elsewhere by this author. For a discussion of its economic significance, see Batten (2000).

friction that prevents a sandpile from collapsing completely to a flat state. It may even be responsible for a special kind of dynamic equilibrium. No doubt you're thinking to yourself: 'Human agents can think but grains of sand can't think! Surely spatial economics must be more sophisticated than sandpiles!' But before we delve more deeply into the quirks and foibles of economic agents, let's explore a few of the surprising features of "unthinking" sandpiles.

Try the following experiment in your backyard sandpit. Starting from scratch on a flat base, build up a pile by randomly adding sand at the centre; slowly and carefully, a few grains at a time. Notice how the grains tend to stick together. The peaked landscape formed by the sand doesn't revert automatically to a flat state when you stop adding sand. A kind of static friction keeps the pile together. Gradually it becomes steeper. Then a few small sand slides start to occur. One grain lands on top of others and topples to a lower level, causing a few other grains to topple after it. In other words, that single grain of sand can cause a local disturbance, but nothing dramatic happens to the pile as a whole.

At this formative stage, events in one part of the pile have no effect on other grains in more distant parts of the pile. The pile is only weakly-interactive, featuring local disturbances between individual grains of sand. As you add more grains and the slope increases, however, a single grain is more likely to cause a larger number of others to topple. If you've created it properly, eventually the slope of your pile will reach a stationary state: the amount of sand you add will be balanced on average by the amount falling off.

There's something unique about this stationary state. Remember that you're adding sand to the pile in the centre, but the sand that's falling off is at the edges. For this to happen, there must be communication between grains at the centre and grains at the edge. How could grains of sand communicate with each other? What transforms this collection of grains from a weakly-interactive to a strongly-interactive pile? Like the sudden appearance of that giant cluster of linkages in our toy example above, suddenly there's communication throughout the entire pile. In the words of Per Bak (1996), the sandpile has attained a *self-organized critical state*.

What might the parallels with a network economy be? Think of the sandpile in its weakly-interactive state as a primitive, autarkic economy. A good image of this kind of economy would be Medieval Europe in the Carolingian era. Carolingian society consisted of villages, and groups of villages, which were mostly self-contained. Villagers were contained within castle walls and self-sufficient manors, mostly making do with the fruits of their surrounding land and forests. Given the risks of travel by land and sea, there would not have been much opportunity to exchange goods over longer distances. At best, Carolingian Europe was a weakly-interactive economic system.

But this isolated state could not remain forever. At a time when long-distance commerce was insignificant and money still a rarity, suddenly the circulation of goods and merchants intensified. All forms of trade rose significantly, but that over longer distances grew most of all. Many larger towns grew suddenly and explosively. In a relatively short period, Europe was transformed from its autarkic, weakly-interactive state into a much more highly-interactive economy. This escalation in trade over longer distances is thought to have been triggered by the opening up of key transportation routes which made travel over longer distances safer. For all intents and purposes, the European economy self-organized.

The marvelous thing about self-organization is that it can transform a seemingly incoherent system into an ordered, coherent whole. Weakly-related grains of sand can be transformed into a strongly-interactive sandpile. Weakly-related human settlements can be transformed into a strongly-interactive merchant economy. In the latter case, freeing up a few critical transport links at a crucial stage transformed the system from a state in which the individual settlements follow their own local dynamics to a self-organized critical state where the emergent dynamics are global. This is a *non-equilibrium phase transition*. Space scales are not merely microscopic, suddenly they're macroscopic. A new organizing mechanism – no longer confined to local interactions – has taken over.

This kind of emergent order seems to be the work of an “invisible choreographer.” (Kauffman 1995). An ordered pattern has sprung up from nowhere. Order through fluctuations, if you like. Technically speaking, the critical state is an attractor for the dynamics. It's a dynamic equilibrium.

Our conclusion is that self-organization can transform disordered, incoherent systems into ordered, coherent wholes. What's amazing is that each emergent whole could not have been anticipated from the properties of the individual elements. Order from incoherence. Who would have thought that a coherent sandpile could result from so many weakly-interactive grains of sand. Who would have thought that a strongly segregated city could result from such weakly sensitive rules about local neighborhood structure?

But that's not all. Once these systems reach a state of spontaneous order, their holistic behaviour seems to follow a dynamic pattern which is lawful in its own right. For example, minor disturbances to a self-organized sandpile can trigger avalanches of all different sizes. The vast majority of these avalanches are small, toppling only a few grains at a time. But a few are much larger. Now and then, an avalanche collapses the entire pile. If we were clever and patient enough, we could measure how many avalanches there are of each size, just like earthquake scientists measure how many earthquakes there are of each magnitude. An interesting thing might happen if we could collect the data and plot the size distribution of avalanches on double logarithmic paper. The likely outcome is shown in Fig. 8.

Surprisingly, the result is a straight line. The x -axis shows the size class, c , to which each avalanche belongs, whereas the y -axis shows how many avalanches, $N(c)$, occurred in that size class. Linearity on a log-log plot confirms that the number of avalanches is given by the simple power law:

$$N(c) = c^{-s}.$$

Taking logarithms of both sides of this equation, we find that

$$\log N(c) = -s \log c$$

Thus the exponent s is nothing more than the slope of the straight line formed when $\log N(c)$ is plotted against $\log c$.

Now reconsider Schelling's segregated city. Chain reactions of relocation – like the sequences of household moves that were triggered by small disturbances to the original, integrated equilibrium – bear a striking resemblance to the avalanches of change depicted in Fig. 8. For starters, the majority of such chain reactions in a city tend to be small in terms of spatial scale. Most



Fig. 8. The size distribution of avalanches in bak's sandpile model conforms to a power law

of them die out locally. But the few larger ones affect a bigger catchment area of residents. Very occasionally, a modest disturbance in a city can trigger a huge chain reaction of responses across the city. Such a skewed size distribution of chain reactions has much in common with the distribution of avalanches underpinning sandpiles, earthquakes, and many other natural phenomena. If we were to collect the data or compute the possibilities exhaustively, the size distribution of chain reactions in our chessboard city would surely obey a power law distribution. This modest task could be undertaken within a relatively small research project. Once again, the aggregate pattern of potential moves may be lawful in its own right.

There's another reason for suspecting that the size distribution of chain reactions leading to segregation may conform to a power law. Schelling's chessboard city, together with his rules determining moves to other locations, correspond to a two-dimensional cellular automaton (CA). CA were originally put into practice by John von Neumann (1966) to mimic the behaviour of complex, spatially extended structures. Because they're really cellular computers, today they're being put to use as simulators, designed to help with time-consuming calculations by taking advantage of fast parallel processing. In Sect. 7, we discuss various examples of CA that have been used to sharpen our intuition about socio-economic behaviour in cities. Schelling's model is not strictly a CA, since it allows agents to migrate from one cell to another.

Since CA employ repetitive application of fixed rules, we should expect them to generate self-similar patterns. Indeed, many do produce such patterns. If Schelling had used computer simulation to explore a much larger chessboard city, self-similar patterns of segregation may have even been visible in

his results. Being akin to periodicity on a logarithmic scale, such self-similar patterns would conform to power laws.

The footprints of power laws can be found everywhere. They turn up in the frequency distribution of many catastrophic events – like floods, forest fires and earthquakes. They’re also thought to be responsible for pink noise, and the music most listeners like best – a succession of notes that’s neither too predictable nor too surprising. In each case, the activity going on is relatively predictable for quite long periods. Suddenly this quiescent state is interrupted by brief and tumultuous periods of major activity, roaming and changing everything along the way. Such punctuations are another hallmark of self-organized criticality.

Although it’s too early to say for sure, it’s likely that many dynamic phenomena generated by human interactions obey power laws.¹² Power laws and punctuated equilibria imply scale invariance, and scale invariance means that no kinks appear anywhere on log-log plots. Economic change may be rife with scale invariance. Nearly 100 years ago, the Italian economist, Vilfredo Pareto, found that the number of people whose personal incomes exceed a large value follows a simple power law.¹³ In some socio-economic contexts, of course, linearity may break down at the smaller and larger scales. The fact that scaling usually has limits does no harm to the usefulness of thinking “self-similar.” In the next section, we’ll look more closely at the role of power laws in the evolution of spatial hierarchies of human settlements.

6. Power laws and the rank-size rule

Most spatial scientists know what happens when we plot the size distribution of all the settlements in the United States against their ranking on double logarithmic paper. Figure 9 shows the results for a span of two hundred years from 1790 onwards. We have ranked the cities in descending order, beginning with the most populated and ending with the least populated. It’s hardly a surprise to learn that each of these downward-sloping curves is pretty close to a straight line. The U.S. system of cities has always conformed roughly to a *rank-size rule*: the population of each city is inversely proportional to its rank.

It is also well known that Zipf (1949) produced dozens of plots of this kind in the 1940s, finding the same kind of regularity.¹⁴ Subsequent research by Berry (1961) showed that macroscopic order also holds for many systems of cities outside the United States. But not every nation conforms to the rule. Some countries display a primate pattern, meaning that the first-ranking city may be much bigger than twice the size of the second-ranking city. For example, the Argentinian, English and French distributions are not rank-size.

¹² In a delightful book about fractals, chaos and power laws, Manfred Schroeder reviews the abundance and significance of power laws in nature and human life; see Schroeder (1991).

¹³ See Pareto (1896).

¹⁴ The rank-size rule is sometimes called Zipf’s Law, in recognition of his observation of regularities in systems of human origin. But Zipf’s discovery was predated by Felix Auerbach (1913), whose German publication is rarely recognized by the English-speaking world. Others to have beaten Zipf to the rule include Lotka in 1925, Gibrat in 1931 and Singer in 1936. Singer (1936) stressed that Gibrat’s “law of proportional effect” is more general, and that the rank-size rule is perfectly analogous to Pareto’s law of income distribution.

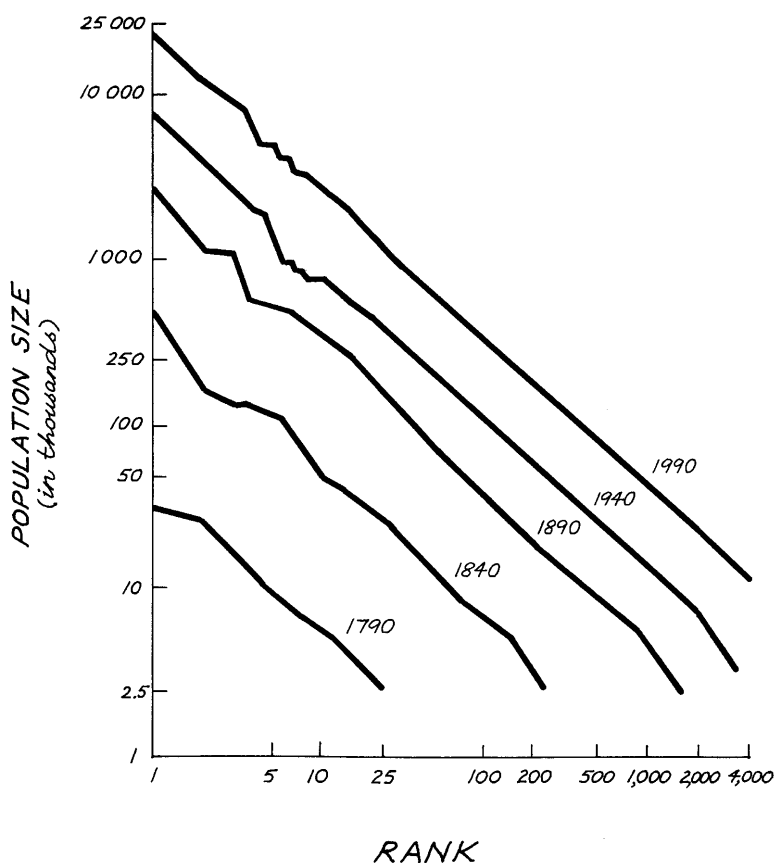


Fig. 9. Rank-size distribution of cities in the United States, 1790–1990

Buenos Aires, London and Paris dwarf their nearest neighbors in their respective distributions. Other nations like Australia display “kinks” or horizontal segments, indicating that some cities are closer in size to each other than the rank-size rule suggests.

Obviously the rule doesn’t hold *perfectly*. In fact it seems to work best in large countries with mostly *self-sufficient* economies, as measured by the ratio of their external to total trade. If this ratio is less than 10%, as it is for the United States and the Soviet Union (now Russia), the rule fits well. It also works well in large countries with long urban traditions – like China and India.

Comparative work suggests that deviations from the rule can often be explained by two factors: (a) improper specification of the complete settlement system, or (b) different qualitative or political stages of development. For example, Portugal’s distribution may be rank-size once it’s recognized that Lisbon heads a larger-than-national urban system.¹⁵ If Singapore and

¹⁵ Portugal has several overseas colonies.

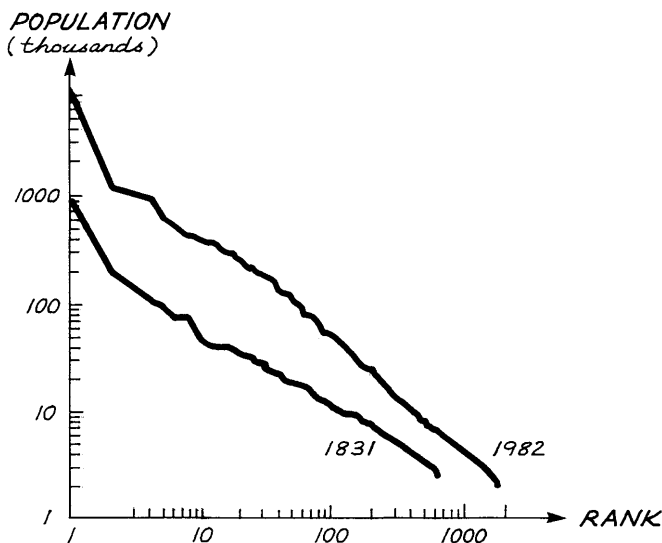


Fig. 10. Rank-size distribution of French settlements, 1831–1982

Malaysia are lumped together – as history demands – then their combined distribution is approximately rank-size (Sendut 1966). The primacy displayed by Japan's city-size distribution also disappears once Tokyo's chief rival is seen to be the multi-centred Kansai conurbation (Batten 1995).¹⁶ These examples show that quite different results may be achieved once cultural or political issues are taken into account. History matters in the world of dynamics.

Since our primary interest is in dynamics, it's fascinating to note just how stable the American rank-size relationship has been for two hundred years. Despite tumultuous changes to its urban system during the last two centuries, Fig. 9 confirms that the rule has applied continuously. Neighboring lines are almost straight and roughly parallel. What incredible stability! Overall growth is depicted by the gradual shift upwards and to the right. A similar story can be found in Europe. There's a high degree of macro-stability in the French urban system, for example, despite the fact that the relative position of individual cities has varied considerably. This state of order is shown in Fig. 10.

What is the explanation for this remarkable stability over time? Could there be something like a universal law of city sizes? Very few urban geographers or economists have tried to answer this intriguing question. Those that have usually argue that it reflects some kind of hierarchy of central places. Perhaps the earliest model linking centres to complementary areas, and thus generating systems of cities consistent with the rank-size rule, was devised by Beckmann (1958). A power law distribution like the rank-size rule is certainly consistent with a simple hierarchy. Yet the hierarchical argument relies on the constancy of some parameters that are unlikely to remain constant over time.

¹⁶ The Kansai or Keihanshin region of Japan includes the cities of Osaka, Kyoto and Kobe, as well as a number of smaller cities. Because of the desire of these cities to foster exchange and co-operate more closely, they appear to be coevolving into a "network city." For a portrayal of the Kansai conurbation's development as a network city, see Batten (1995).

Central place theory is a static theory. How can a static theory possibly hope to explain something that results from a decidedly dynamic chain of events?

Recently, Krugman and others have noted that Simon's (1955) model of random growth can produce a huge range of city sizes whose upper tail is well described by a power law (see Krugman 1996; Fujita et al. 1999). Simon envisaged a process in which urban populations grow over time by discrete increments. A city is simply a "clump of lumps" whose expected growth rate is independent of size. Despite its ability to predict a power law, the assumption that the urban size distribution tends to approach a steady state is too unrealistic, and the model itself too nihilistic, to offer an *explanation* for the rank-size rule.

Berry (1961) also offered an explanation that relied on the steady-state argument. As the economic, political and social life of a country grows more complex, he argued that its city-size distribution will evolve towards a rank-size pattern, because this represents the *steady-state* of the whole urban system. In other words, economic development will move an immature system of cities closer and closer to a rank-size distribution over time. This suggests that there's a scale from primate to rank-size, which is somehow tied to the maturity and complexity of the interactive forces affecting a nation's urban structure. When few strong forces prevail, primacy results. When many strong forces prevail, the rank-size rule results.

In view of what we've learnt earlier about complex adaptive systems, perhaps we can be more explicit about the coevolutionary process involved. The rank-size distribution may be an emergent *attractor* in the phase space of possible dynamics governing urban change. This means that individual towns and cities *self-organize* in order to achieve then preserve this rank-size pattern over time. Indeed there's evidence to support the idea that the rank-size pattern is such an attractor (see Haag and Max 1993). The macro stability of the American and French urban systems has persisted despite the changing position of individual cities in each nation's urban hierarchy. There's no sign of proportionate growth in these systems.

Can we visualize what such a rank-size attractor might look like? A possible metaphor is the upper canopy of a forest, containing trees of various heights and ages. As the different trees mature, the profile (or contour) of the canopy shifts upwards over time, just like the rank-size distributions. But its shape never seems to alter, despite the fact that individual trees change markedly. Some grow quickly, others grow slowly; some die, others are born. Could it be that towns and cities in the U.S. system manage to self-organize into a kind of equilibrium pattern like trees in a forest? If so, then it's a special kind of dynamic equilibrium. We might even think of it as a rank-size *ecology*. While some cities grow and others decline, the overall ecology (i.e. the profile of the aggregate city-size distribution) doesn't change.

Speaking of ecologies, the rank-size distribution looks like a higher level "simplicity" that emerges from the interactive activities linking cities and their coevolution. Such an unexpected result collapses the apparent chaos of a highly-interactive system into a very simple rule. As we've stressed already, the urban hierarchy of the United States is not the only nation to exhibit this kind of macro-stability. A similar order can be found in Asia. Indonesia, Japan, Malaysia-Singapore, South Korea and Taiwan have more-or-less preserved their rank-size distributions over the last fifty years, despite many individual towns and cities "jumping rank" dramatically (Sendut 1966).

The rectilinearity of rank-size plots has been shown to rephrase an underlying scaling distribution. Zipf put forward the bold claim that scaling is the “norm” for all social phenomena. Mandelbrot’s (1997) work has added weight to Zipf’s claim.¹⁷ A special feature of all these coupled dissipative systems is that they may evolve naturally towards a self-organized critical state. Perhaps the rank-size condition corresponds to a state of self-organized criticality, where cities are formed by avalanches of human migration. In such a state, there’s plenty of communication between each and every part of the urban system. Even the most peripheral settlements communicate with the central cities. Just like a self-organizing sandpile.

Zipf reasoned that man’s dual role as a producer and a consumer posed a profound conflict in the economy of his location. Two extreme outcomes could result. One course of action involved moving the people close to the sources of raw materials to save on transporting materials to the people. The effect of this was to split the whole population into a large number of small, widely scattered, autarkical communities, having virtually no communications or trade with one another. Sounds very much like Medieval Europe’s self-sufficient, agrarian economy, doesn’t it?

The other course of action moved the materials to the population. All production and consumption would take place in one big city, where the entire population would live.¹⁸ In practice, of course, neither extreme occurs. The actual location of the population depends on the comparative magnitudes of both forces in question. One force makes for a larger number of communities of smaller size, and the other makes for a smaller number of communities of larger size, so the realized outcome must balance these forces, leading to a rank-size distribution of cities.

What’s fascinating about these opposing forces is that one produces a *simple*, weakly-interactive economy; but the other produces a *complex* economy that’s strongly-interactive. Putting all the population in just one city is certainly an extremely interactive solution, bordering on the chaotic. We might even think of it as a solution that’s *too-strongly-interactive*! Most nations’ settlement systems lie between these two extremes. Some developed economies are more strongly-interactive, whereas other developing economies are only weakly-interactive. Some are dynamically stable, others are potentially unstable.

These two extreme conditions correspond to the ordered and the chaotic. Like our old friend, the sandpile, a self-organized system is poised to unleash an avalanche of small, medium, and large changes throughout an economic system of interacting agents. The result is that a coevolving economy gets driven away from the ordered regime towards the chaotic regime, but soon gets driven right back again. Order to chaos, then chaos to order, forever adaptive. Thus it’s most probable state is somewhere in between. At the edge of chaos, if you like. Our tendencies to simplify and “complexify” are power-

¹⁷ Zipf’s law proved interesting to Benoit Mandelbrot when he was a post-doctoral student at MIT. It propelled Mandelbrot on a path that led to finance and economics, and later to fractals.

¹⁸ Krugman asserts that when Marshallian dynamics are added to the traditional constant-returns, competitive model of international trade, they result in a rather simple landscape – one where the whole space of possible resource allocations drains to a single point; see Krugman (1994), p. 412.

ful forces that shape much of our behaviour, so it's very likely that they shape our spatial landscapes as well.

7. Artificial cities

Urban theory has largely failed in its quest to model and explain such oscillatory patterns of behaviour. Recently, the multifractal dimension of rank-size distributions has been established (see Haag 1994; Mandelbrot 1997). For many urban analysts, however, the rank-size rule remains a quaint curiosity and their understanding of how a city self-organizes over time is modest at best. In this section, we'll explore an alternative means of probing socio-economic dynamics at the urban level: *agent-based simulations*.

Simulation games that deal with urban problems are gaining in popularity. The success of software packages such as SIMCITY are proof of that. In this section, we'll restrict ourselves to a few simulations using cellular automata (CA), because this breed of simulation boasts two advantages over other simulation techniques. First, it's explicitly dynamic. Second, it links macro-structure to microbehaviour. In an earlier section, we stated that Schelling's chessboard model of segregated neighborhoods corresponds to a two-dimensional CA. In the same issue of the *Journal of Mathematical Sociology* that published Schelling's famous article, there's a lesser-known article by James Sakoda (1971) entitled "The Checkerboard Model of Social Interaction." After describing a similar model to Schelling's, Sakoda stresses that the main purpose of cell-based modeling is not a predictive one, but clarification of concepts and "insight into basic principles of behaviour." It's these insights that make CA and checkerboard modeling promising when it comes to deepening our primitive understanding of socio-economic dynamics.

A two-dimensional CA consists of the following: (1) a two-dimensional grid; (2) at each grid site, there's a cell which is in one of a *finite* number of possible states; (3) time advances in *discrete* steps; (4) cells change their states according to *local* rules, so that the state of a cell in the next period depends upon the states of neighboring cells in past periods; (5) the transition rules are mostly *deterministic*, although nondeterministic rules are also possible; (6) the

Table 2. Similarities between CAs and socio-economic dynamics

	Cellular automata	Socio-economic dynamics
Basic elements	Cells are the basic units or "atoms" of a CA	Individual agents are the basic units of an economy
Possible states	Cells assume one of a set of alternative states	Agents form mental models which enable them to make choices from alternatives
Interdependence	The state of a cell affects the state of its closest neighbors	The choices made by an agent affect the choices made by other agents
Applications and tasks	Modeling the emergence of order, macro outcomes explained by micro rules, and the path dependence of dynamic processes	Important tasks include: understanding the emergence of order, macro to micro relationships, and economic dynamics

system is *homogeneous* in the sense that the set of possible states is the same for each cell and the same transition rule applies to each cell; (7) the updating procedure usually consists of applying the transition rule *synchronously* or selecting cells *randomly*.¹⁹

The number of different transition states in a CA can quickly go through the roof. Consider a two-dimensional CA with just two possible cell states, a neighborhood of one cell and its four orthogonally adjacent neighbors, and with only the last period having an influence on the next period. In such a seemingly simple case, the number of different transition states is $2^{32} = 4,294,967,296$! No wonder we need a computer to implement a CA-based approach to simulation. The fortunate thing is that the kinds of problems tackled successfully in some of the physical sciences using CA just happen to be amongst the most urgent, unsolved problems in the social sciences. Economics is a case in point. Table 2 provides a comparative overview.

Using the simplest CA, it's easy to show that complex global patterns can emerge from the application of local rules. Schelling's patterns of segregation are an example of global emergence, and emergence is one of the things that makes CAs so intriguing. In a world where global outcomes fuse in subtle and diverse ways with local action, CAs look like a methodological paradigm for the 21st century.²⁰ They're the source of, and inspiration for, major developments in complex adaptive systems. The promising new field of Artificial Life is one of the more obvious examples. The message is that many classes of dynamics can be simulated through CA.

Perhaps the greatest attraction of a CA-based approach to socio-economic dynamics is the equal weight given to the importance of space, time and system attributes. When Sakoda and Schelling published their checkerboard articles, however, they didn't mention the CA concept. They must have been blissfully unaware of it. Yet it's clear that CA and socio-economic dynamics have a great deal in common (as Table 2 testifies). Checkerboard models also share some obvious features with CA, like grid structure and local neighborhoods. At the same time, Sakoda's and Schelling's checkerboard models focus primarily on "sorting and mixing," i.e. agents searching for and moving to attractive locations in space. Checkerboard models don't just concentrate on cells changing their state at a given site (like CAs), but on changing their site as well. For simulations over geographical space, this point is important. We must distinguish between models that allow individuals to move – *migration* models – and those that do not – *steady site* models (Hegselmann 1996). Moving an agent to an empty cell in his neighborhood can be treated as the application of a rule by which an occupied cell and a neighboring empty one exchange states.

Another important feature of CAs is the definition of neighborhoods. Two kinds are popular in two-dimensional CA: the *von Neumann* neighborhood

¹⁹ Although von Neumann and Stanislaw Ulam were the first to introduce the CA concept about fifty years ago, it's pretty safe to say that John Conway popularized the concept through his invention of the game of "Life." In Life, cells come alive (i.e. "turn on"), stay alive (i.e. "stay on") or die (i.e. "turn off"), depending on the states of neighboring cells. Although Life is the best known CA, it's perhaps the least applicable to real configurations.

²⁰ In their introduction to a special issue of the journal *Environment and Planning B* devoted to urban systems as CA, Batty et al. (1997) make this suggestion. The reader is directed to this issue for an overview of ways in which urban dynamics can be simulated through CA.

(four neighboring cells north, south, east and west of the cell in question) and the *Moore* neighborhood (with the same four cells plus those which are NW, NE, SE and SW). More distant neighbors may have an influence on state changes, but it's assumed in strict CA that the temporal dynamics will take care of these effects. In other words, growth and decline imply spatial diffusion. Fortunately, a halfway house exists, embracing some CA principles but also relaxing the neighborhood definition. These are the so-called cell-space (CS) models introduced by Albin (1975).

Although many CA applications reported in the urban modeling literature relax the neighborhood effect to allow for action-at-a-distance, this is not in the spirit of strict CA. Perhaps the most important challenge for this class of simulation model is the specification of the nexus between urban changes at the physical and the human levels. Two key systems are of interest: (1) the relatively slow developmental changes that take place across networks of housing and infrastructure constructed in cities, and (2) the relatively rapid behavioural changes that agents can implement by altering their own mental models and choices. On the surface at least, one might argue that CA transition rules should be based on agents' local behaviour. However, real urban "cells" like houses, roads and green areas are more spatial in character, and are governed by a broader set of coevolutionary forces. The real challenge is to address the tangible and intangible changes interdependently.

Recent studies have demonstrated the self-organizing properties of some urban simulation models. For example, a simple heuristic CA model, called *City*, was developed to study sociospatial segregation in a similar spirit to Schelling's work (see Portugali et al. 1994). *City*'s territory is a two-dimensional square lattice of cells, each of which may be regarded as a house or a *place*. Individuals (persons, families or households) occupy or leave various places, thereby generating the migration dynamics and sociospatial structure of the *City*. Residents and place-hunters base their decisions on preferences about the types of individuals in neighboring places. Model results display self-organization, local instabilities, captivity, and other interesting phenomena. Later versions feature two-levels: a population level composed of individuals with cultural and economic properties, and a housing-stock level consisting of a two-dimensional lattice of cells. Immigrants and inhabitants interact with each other and the system of cells (houses), and this interaction gives rise to migration dynamics, changes in the properties of individuals, and changes in the properties of cells (Portugali et al. 1997).

Another interesting multiagent-based model goes by the name of *SIM-POP*. It was developed by a French group of social scientists, with the aim of unearthing a set of rules which transform systems of cities over time (see Sanders et al. 1997). They experiment with the effects of various hypotheses using a grid of hexagonal cells. Settlements are characterised by types of economic functions. The general evolutionary patterns which emerge from their work are consistent with the arguments put forward elsewhere in this paper. For example, the universality of power laws and the rank-size rule is demonstrated under a variety of initial conditions. Their model also suggests that transitions between different urban regimes are a necessary characteristic of urban evolution.

One of the strengths of agent-based simulation is the ability to generate simple and complex regimes of behaviour. CA-based systems can simplify and complexify life in its various forms. Cities are a perfect illustration of this.

Urban cells typically switch from being weakly-interactive to states in which they're *too-strongly-interactive*. If the pendulum swings too far too quickly, collective outcomes can be counterproductive. Congestion and pollution are typical examples. This type of oscillatory behaviour is an important component of urban evolution, and will be discussed more fully in the next section.

8. Growing a silicon society

In this section, we discuss an interesting, two-level, silicon world, the joint brainchild of Joshua Epstein and Robert Axtell (1996). They set out to “grow” a social order from scratch, by creating an ever-changing environment and a set of agents who interact with each other and their environment according to a set of behavioural rules. History is said to be an experience that's only run once. Clearly Epstein and Axtell don't hold with that view. Their idea is that an entire society – like a spatial economy complete with its own production, trade and culture – could be “recreated” from the interactions among the agents. As Epstein suggests: “You don't solve it, you evolve it.” They call the laboratory in which they conduct their simulation experiments a *CompuTerrarium*, and the landscape which the interacting agents inhabit a *Sugarscape*.²¹ Let's take a closer look at how socio-economic life develops in this artificial world.

The action takes place on a grid of fifty-by-fifty cells. But the landscape denoted by this grid is not blank, as it is on a typical CA. On it is scattered this silicon world's only resource: sugar. In order to survive, the entities that inhabit this sweetened landscape must find and eat the sugar. The entities themselves are not just cells that are turned on or off, mimicking life or death. Each is an agent that's imbued with a variety of attributes and abilities. Epstein and Axtell call these *internal states* and *behavioural rules*. Some states are fixed for the agent's life, while others change through interaction with other agents or with the environment. For example, an agent's sex, metabolic rate, and vision are hard-wired for life. But individual preferences, wealth, cultural identity and health can all change as agents move around and interact.

Although every interacting agent appears on the grid as a colored dot, each may be quite different. Some are far-sighted, spotting sugar from afar. Others are thrifty, burning the sugar they eat so slowly that each meal lasts an eternity. Still others are short-sighted or wasteful. Rapacious consumers eat their sugar too quickly. The obvious advantage of this heterogeneity is that it's capable of mimicking (albeit simplistically) the rich diversity of human populations in terms of their preferences and physiological needs. Any agent that can't find enough sugar to sustain its search must face that ultimate equilibrium state: it simply dies!

Sugarscape resembles a traditional CA in its retention of rules. There are rules of behaviour for the agents and for the environmental sites (i.e. the cells) which they occupy. Rules are kept simple, and may be no more than the commonsense ones for survival and reproduction. For example, a simple movement rule might be: *Look around as far as you can – find the nearest*

²¹ Readable summaries of this metaphoric world of artificial life can be found in Casti (1997, Chap. 4) and Ward (1999, Chap. 2).

location containing sugar – go there – eat as much as you need to maintain your metabolism – save the rest. Epstein and Axtell speak of this as an agent-environment rule. A rule for reproduction might be: *Breed only if you've accumulated sufficient energy and sugar.* Also, there are rules governing socio-economic behaviour, such as: *Retain your current cultural identity (e.g. consumer preferences) unless you see that you're surrounded by many agents of a different kind – if you are, change your identity to fit in with your neighbors or try to find a culture like your own.* This rule smacks of Schelling's segregation model, because it highlights coevolutionary possibilities among nearby neighbors.

The *CompuTerrarium* leaps into action when hundreds of agents are unleashed randomly onto the grid. Colored dots distinguish agents who can spy sugar easily from more myopic agents. Naturally, all the agents rush towards the sugar. The latter may be piled into two or more huge heaps or scattered more evenly throughout the landscape. Strikingly, many agents tend to "stick" to their own terrace, adjacent to their "birthplace." Because natural selection tends to favour those agents with good eyesight and a low metabolic rate, they survive and prosper at the expense of the short-sighted, rapacious consumers. In short, the ecological principle of carrying capacity quickly becomes evident. Soon the landscape is covered entirely with red dots (high-vision agents).

Even with relatively simple rules, fascinating things happen as soon as the agents begin to interact on the Sugarscape. For example, when seasons are introduced and sugar concentrations change periodically over time, high-vision agents migrate. But low vision, low metabolism agents prefer to hibernate. Agents with low vision and high-metabolism usually die, because they're selected against.

All of the time, the surviving artificial agents are accumulating wealth (i.e. sugar). Thus there's an emergent wealth distribution on the Sugarscape. Herein lies the first topic of particular interest to us. Will the overall wealth be distributed equally, or will agents self-organize into a Pareto distribution? In other words, will equity prevail or will the ubiquity of power laws prevail again? Although quite symmetrical at the start, the wealth histogram on the Sugarscape ends up highly skewed. Because such skewed distributions turn up under a wide range of agent and environment conditions, they resemble an emergent structure – a stable macroscopic pattern induced by the local interaction of agents. Self-organization seems to be on the job as usual, and the power law prevails yet again!

Although these few examples are a useful way of illustrating the variety of artificial life evolvable on the Sugarscape, they hardly herald an impending revolution in our understanding of how a spatial economy works. For that we must expand the behavioural repertoire of our agents, allowing us to study more complex socio-economic phenomena. Epstein and Axtell have made a start on this expansion. When a second commodity, spice, is added to the landscape, a primitive trading economy emerges. By portraying trade as welfare-improving barter between agents, they implement a trading rule of the form: *Look around for a neighbor with a commodity you desire, bargain with that neighbor until you agree on a mutually acceptable price, then make an exchange if both of you will be better off.*

Surprisingly, this primitive exchange economy allows us to test the credentials of that classical theory of market behaviour: the efficient market hy-

pothesis. The first stage of the test involves imbuing agents with attributes consistent with neoclassical economic wisdom – homogeneous preferences and infinite lifespans for processing information. Under these conditions, an equilibrium price is approached. But this equilibrium is not the general equilibrium price of neoclassical theory. It's *statistical* in nature. Furthermore, the resulting resource allocations, though locally optimal, don't deliver the expected global optimum. There remain additional gains from trade that the agents can't extract. What we find is that two competing processes – exchange and production – yield an economy that's perpetually out of equilibrium.

Once we imbue agents with human qualities – like finite lives, the ability to reproduce sexually, and the ability to change preferences, the trading price never settles down to a single level. It keeps swinging between highs and lows, very much like price oscillations in real markets. Basically, it appears to be a random distribution. But it turns out that there's structure after all! Although the seemingly-random price fluctuations continue indefinitely, the fluctuations appear to be variations from an identifiable price level. This particular price just happens to be the same equilibrium level as the one attained under those all-too-unrealistic assumptions underpinning the efficient market hypothesis. Thus we gain the distinct impression that any equilibrium state associated with the efficient market hypothesis is nothing more than a limiting case among a rich panorama of possible states that may arise in the marketplace. As Epstein suggests: "If the agents aren't textbook agents – if they look a little bit human – there is no reason to assume markets will perform the way economic textbooks tell us they should."

How is the distribution of wealth affected by trade? It turns out that the overall effect of trade is to further skew the Sugarscape's distribution of wealth. By increasing the carrying capacity, and allowing more agents to survive, it also magnifies differences in wealth. Trade increases the interactions between agents, thereby strengthening the power law fit even further. Experiments with a wider set of choice possibilities endorsed the view that it's devilishly difficult to find conditions under which a society's wealth ends up being evenly distributed. We may conclude that there's a definite tradeoff between economic equality and economic performance. This bears a striking qualitative similarity to findings in various economies around the world.

There's so much more one could say about the socio-economic laboratory constructed by Epstein and Axtell. Many other issues – such as the emergence of cultural groups, webs of economic intercourse, social clusters, institutional structures, and disease – can all be scrutinized under the Sugarscape microscope. The pair are now working to extend the Sugarscape in order to capture the way of human life in the late twentieth century. Thus far, the agents have sex but there are no families, no cities, no firms and no government. Over the next few years, they hope to produce conditions under which all of these things emerge spontaneously. As life on the Sugarscape is in its infancy at present, who's to say what might happen in time?

Sugarscape is an important example of agent-based simulation for several reasons. First, although economists and other social scientists study society, they do so in isolation. Economists, geographers, psychologists and sociologists rarely interact meaningfully or pool the knowledge they've accumulated. Regional scientists are more cooperative, but the current organization of university departments further endorses these divides. Yet life on the Sugarscape brings all these narrow views together, broadening our understanding in a

meaningful way. Second, Sugarscape activities are interactive and dynamic. Thus it's far more process-dependent than classical models. Third, Sugarscape recognizes and preserves differences in culture and skills that human populations exhibit. Finally, for the first time in history, the social sciences have the opportunity to conduct and repeat experiments and test hypotheses to do with social and economic behaviour. Sugarscape typifies this new way of doing social science as we enter the unprecedented era of agent-based simulation. If your business is modelling socio-economic behaviour over space, it's an excellent starting point for rule-based experiments.

Simulation models like *City*, *SIMPOP* and Sugarscape have shown us that complex behaviour need not have complex origins. Some of the complex behaviour arises when agents are imbued with relatively simple predictors or behaviour generators. Other emergent behaviour is attributable to predictors which differ in terms of the time horizons over which they're applied. Since it's hard to work backwards from a complex outcome to its generator(s), but far simpler to create many different generators and thus synthesize complex behaviour, a promising approach to the study of complex socio-economic systems is to undertake a general study of the kinds of collective outcomes that can emerge from different sets of predictors as behaviour generators (see Rasmussen and Barrett 1995; Barrett et al. 1998). As we've stressed already, work of this kind must be done by simulation experiments.

9. Some final thoughts

The real difficulty with the spatial economy is that each of us is part of the very thing that we're desperately trying to understand. This has the hallmark of a systems problem. But it's not a classical systems problem, like how a clock "tells the time" or how a car "moves".²² Clocks and cars are structurally complex, but behaviourally simple. Their behavioural simplicity transcends the structural complexity of all their intricate parts. A spatial economy, however, is behaviourally complex. Because some "parts" are human agents, they're observers as well as participants, learning from their experiences as well as contributing to the collective outcome. What people believe affects what happens to the economy *and* what happens to the economy affects what people believe. Thus any serious study of emergence must confront *learning* (see Holland 1998; Batten 2000).

Whenever agents learn from, and react to, the moves of other agents, predicting the collective outcome is only possible when the economy is linear. If learning is only weakly-interactive, the behaviour of the whole economy is just the sum of the behaviour of its constituent parts. For learning to be adaptive, however, the stimulus situations must themselves be steadily evolving rather than merely repeating. This requires strongly-interactive conditions. The existence of a recursive, nonlinear feedback loop is the signature of coevolutionary learning. People learn and adapt in response to their *recent* experiences. Each agent's decision affects other agents, and thus the collective outcome as a whole; and this collective outcome, in turn, influences the

²² As Cohen and Stewart have noted, "You can dissect axles and gears out of a car but you will never dissect out a tiny piece of motion"; see Cohen and Stewart (1994, p. 169).

agents' future beliefs and decisions. In other words, the behaviour of the whole is greater than the sum of its parts.

Unexpected outcomes trigger avalanches of uncertainty, causing each agent to modify his view of the world. How the world looks to each of us depends on the kind of "glasses" we're wearing. As Kant suggested, nobody can have certain knowledge of things "in themselves." Each of us only knows how things appear to us. If we're only privy to part of the information about a spatial economy, then there are clear limits to what we can know. Each agent's mind sets these limits. When we ask questions about this economy, we're asking about a totality of which we're but a small part. We can never know such an economy completely; nor can we see into the minds of all its agents and their idiosyncracies.

The key to understanding adaptive behaviour lies with explanation rather than prediction. When economic agents interact, when they must think about what other agents may or may not be thinking, their collective behaviour can take a variety of forms. Sometimes it might look chaotic, sometimes it might appear to be ordered, but more often than not it will lie somewhere in between. At one end of the spectrum, chaotic behaviour would correspond to *rapidly* changing models of other agents' beliefs. If beliefs change too quickly, however, there may be no clear pattern at all. Such a volatile state could simply appear to be random. At the other end of the spectrum, ordered behaviour could emerge, but only if the ocean of beliefs happens to converge onto a mutually consistent set of models of one another.

For most of the time, however, we'd expect that mental models of each other's beliefs would lie somewhere in between these two extremes, tending to change, poised ready to unleash avalanches of small and large changes throughout a system of interacting agents. Why should we expect this? Given more data, we would expect each agent to improve his ability to generalize about the other agents' behaviour by constructing *more complex* models of their behaviour. These more complex models would also be more sensitive to small alterations in the other agents' behaviour. Thus as agents develop more complex models to predict better, the coevolving system of agents tends to be driven away from the ordered regime toward the chaotic regime. Near the chaotic regime, however, such complexity and changeability would leave each agent with very little reliable data about the other agents' behaviour. Thus they would be forced to simplify, to build *less complex* models of the other agents' behaviour. These less complex models are less sensitive to the behaviour of others and live in calmer oceans.

Because it's impossible to formulate a closed-form model under such volatile conditions, traditional models of the spatial economy fail in this environment. A typical model of a spatial economy is an attempt to gain understanding of (parts of) that economy through a simplified representation of (parts of) it. If that economy is a complex adaptive system, however, the set of unknown predictors is too large and variable to be simplified in a meaningful way. In John Holland's jargon, the full set of predictors forms an *ecology*. If we want to understand how this ecology might evolve over time, we're forced to resort to simulation experiments. Simulation doesn't simplify that economy, but incorporates as much detail as is necessary to produce emergent behaviour. There's simply no other way of accommodating such a large, ever-changing population of active predictors.

The defining characteristic of a complex adaptive system like a spatial

economy is that some of its global behaviours cannot be predicted readily from knowledge of the underlying interactions (Darley 1995). Instead, this kind of behaviour is emergent. An emergent phenomenon is defined as collective behaviour which doesn't seem to have any clear explanation in terms of its microscopic parts. What does emergence tell us? It tells us that an economic system of interacting agents can spontaneously develop collective properties that are not at all obvious from our limited knowledge of each of the agents themselves. These statistical regularities are large-scale features that emerge purely from the microdynamics. They signify order despite change. Sometimes, this order takes the form of self-similarity at different scales.

For emergent phenomena, the optimal means of "modelling" is simulation (Darley 1995). Simulation is really a blend of modelling and computation. It's basically the art of using computers to calculate the interactions among separate algorithmic representations. In a spatial economy, for example, the algorithmic representations might be agents (firms) engaged in the business of trade between regions. If developed in an appropriate fashion, at the very least a simulation should demonstrate sufficient understanding of the original economic system so as to be able to reproduce its behaviour. But surely if we understand something very well, we shouldn't need to perform such a simulation. Sadly, the complexities within a spatial economy may preclude such a deeper understanding.

Our astonishment at the fact that we seem unable to predict emergent properties doesn't stem from any inability to understand, but from those inherent properties of the system attributable to the spatial scale of interactions. As systems become more emergent, the propagation of information through accumulated interaction will blur the boundaries of any analysis which we try to perform. All useful predictive knowledge is contained in the accumulation of interactions.

A major advantage of an agent-based simulation is that the system's dynamics is generated by way of the simulation. Interactions can accumulate, multiple pathways can be recognized, and emergent properties can be revealed, all without making any ad hoc assumptions about these properties. The major disadvantages of such a simulation are the extremely high computational demands and the fact that it may not always lead to a better understanding of the basic mechanisms that caused the dynamics. It reveals, but doesn't always explain, the inherent dynamics.

Perhaps it's time for a synthetic approach to spatial economics. Instead of taking economies apart, piece-by-piece, spatial economics could attempt to put them together in a coevolutionary environment. We might even find that a synthetic approach may lead us beyond known phenomena: beyond *economic-life-as-we-know-it* and into the less familiar world of *economic-life-as-it-could-be*. It would address the problem of creating diverse behaviour generators, which is partly psychological and partly computational. Like nature, a spatial economy is fundamentally parallel. Thus we must recapture economic life as if it's *fundamentally and massively parallel*.²³ If our models are to be true to economic life, they must also be highly distributed and massively parallel.

There are some exciting experiments underway which attempt to replicate the rich diversity of socio-economic life inside the computer. We've discussed

²³ Massively parallel "architecture" means that living systems consist of many millions of parts, each one of which has its own behavioural repertoire.

a few of these in this paper. Their common feature is that the main behaviours of interest are properties of the interactions between agents, rather than the agents themselves. The *virtual parts* of an economy depend on nonlinear interactions between human agents for their very existence. If we choose to isolate the agents, then the virtual parts disappear. If we choose to aggregate the agents, then the virtual parts disappear. It's the virtual parts of an economy that the new spatial economics must seek. In this quest, synthesis through simulation is a key methodological tool and the computer can serve as the scientific laboratory.

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